

Member ID: _____

Time: _____

Rank: _____



COMPUTER NETWORK TECHNOLOGY (300)

STATE 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each) _____ (100 points)

TOTAL POINTS _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

MULTIPLE CHOICE

Identify the letter of the choice that *best* completes the statement or answers the question.

1. Which of the following is a valid DHCP operation that occurs after a client broadcast?
 - a. Repeat
 - b. Assess
 - c. Obtain
 - d. Discover
2. Which of the following is **false** regarding PXE?
 - a. Stands for Preboot eXecution Environment
 - b. Allows booting to an operating system over a network connection
 - c. Requires a direct physical connection between the client and server
 - d. Utilizes DHCP options 66 and 67 to advertise the PXE boot server
3. Which of the following is **true** regarding CNAME records?
 - a. Resolves hostnames to IP addresses
 - b. Maps an alias to the canonical name
 - c. Resolves IP address to hostnames
 - d. Specifies a port for specific services
4. Which of the following is a valid host address?
 - a. 10.127.255.255/9
 - b. 172.16.63.255/18
 - c. 192.168.0.30/27
 - d. All of these are correct
5. Which of the following accurately describes a SSID?
 - a. Stands for Service Set Identifier
 - b. Unique name that identifies a wireless network
 - c. Can be 32 characters in length
 - d. All of these are correct
6. Which of the following problem descriptions would most likely be caused by a DNS issue?
 - a. A server can be pinged by IP address, but not by hostname
 - b. A server can be pinged by hostname, but not IP address
 - c. A server can resolve external websites to an IP address, but cannot ping the IP address
 - d. None of these are correct

7. Which of the following best describes DHCP snooping?
 - a. Excludes rogue DHCP servers and removes malicious / malformed DHCP traffic
 - b. Inspects and monitors DHCP traffic, but does not interfere
 - c. Captures and forwards all DHCP traffic to the DHCP sanitation server
 - d. Penetrates security vulnerabilities in an existing DHCP infrastructure
8. You are asked to troubleshoot an issue on a domain controller. The issue is that the time is drifting (the reported time is gradually deviating from an accurate time source). Which of the following protocols is the most likely culprit?
 - a. PST
 - b. NTP
 - c. UTC
 - d. GMT
9. Which of the following describes what the abbreviation “MU-MIMO” stands for?
 - a. Multi-Unit, Multiple-Ingress, Multiple-Obfuscation
 - b. Multi-Upload, Multiple-Inbound, Multiple-Outbound
 - c. Multi-Unload, Multiple-Instance, Multiple-Overwrite
 - d. Multi-User, Multiple-Input, Multiple-Output
10. Which of the following best describes how a server would connect to a DAS array?
 - a. Over the network, via iSCSI
 - b. Direct connection, via SAS
 - c. Over the network, via SMB
 - d. Direct connection, via Bluetooth
11. Which of the following best describes how a server would connect to a NAS array?
 - a. Over the network, via iSCSI
 - b. Direct connection, via SAS
 - c. Over the network, via SMB
 - d. Direct connection, via Bluetooth
12. Which of the following best describes how a server would connect to a SAN array?
 - a. Over the network, via iSCSI
 - b. Direct connection, via SAS
 - c. Over the network, via SMB
 - d. Direct connection, via Bluetooth
13. Which of the following describes a static IP address?
 - a. An IP address that manually assigned
 - b. An IP address that is assigned by a DHCP server
 - c. An IP address that is assigned when a DHCP server cannot be reached
 - d. An IP address that is assigned as the default gateway

14. Which of the following describes an APIPA address?
- a. An IP address that manually assigned
 - b. An IP address that is assigned by a DHCP server
 - c. An IP address that is assigned when a DHCP server cannot be reached
 - d. An IP address that is assigned as the default gateway
15. What do double colons indicate regarding IPv6 addresses?
- a. Separation between the prefix and suffix
 - b. Separation between the MAC address and IP address
 - c. Replaces a consecutive series of the letter "F"
 - d. Replaces a consecutive series of zeroes
16. Which of the following is a valid IPv4 or IPv6 loopback address?
- a. 127.0.0.1
 - b. ::1
 - c. 0:0:0:0:0:0:1
 - d. All of these are correct
17. Which of the following is the **last** usable host address in the following network:
192.168.0.128/26?
- a. 192.168.0.190
 - b. 192.168.0.254
 - c. 192.168.0.158
 - d. None of these are correct
18. Which of the following is the subnet mask for CIDR notation /17?
- a. 255.254.0.0
 - b. 255.255.128.0
 - c. 255.255.192.0
 - d. 255.255.255.224
19. Which of the following is the correct CIDR notation for the 255.248.0.0 subnet mask?
- a. /26
 - b. /19
 - c. /13
 - d. /11
20. Which of the following describes a DHCP exclusion range?
- a. IP addresses within the range are unleaseable
 - b. IP addresses within the range must be assigned statically
 - c. Can be used to exclude a single IP address or a range
 - d. All of these are correct

21. Which of the following environments would you most likely see a Star network topology?
- a. Internet
 - b. Datacenter
 - c. Residential
 - d. None of these are correct
22. Which of the following network topologies would be used to describe the Internet?
- a. Star
 - b. Tree
 - c. Ring
 - d. Mesh
23. Which of the following describes network baselining?
- a. Resetting all network devices to factory defaults
 - b. Measuring your network over a period of time to obtain performance metrics
 - c. Applying vendor recommended best practices and updates
 - d. Ensuring that all network equipment is from the same vendor
24. Which of the following would most likely be the cause of a broadcast storm?
- a. Spanning tree protocol
 - b. Switching loop
 - c. Link aggregation
 - d. All of these are correct
25. Which of the following describes a Standard Access Control List?
- a. A list of permit or deny rules by source IP address
 - b. A list of whitelisted domains
 - c. A list of blacklisted domains
 - d. A list of permit or deny rules by source and destination IP addresses
26. Which of the following would be most helpful with regards to network security?
- a. Single Factor Authentication
 - b. DMZ
 - c. BYOD
 - d. Firewall
27. Which of the following is **not** a common type of social engineering attack?
- a. Phishing
 - b. Tailgating
 - c. DDoS
 - d. Pretexting

28. Which of the following is **not** a valid benefit or use for a proxy server?
- a. Content-filtering
 - b. Bypassing filters
 - c. Improved security
 - d. Network address translation
29. As a network administrator, you are asked to troubleshoot a client connectivity issue. The issue is that a specific client cannot reach the internal website. All other clients can reach the internal website without issue. Given the following potential issues, which is the most likely culprit?
- a. The client firewall is blocking port 443
 - b. The server firewall is blocking port 443
 - c. The client default gateway is incorrect
 - d. The server default gateway is incorrect
30. Which of the following is **not** a layer of the TCP/IP model?
- a. Application
 - b. Transport
 - c. Data Link
 - d. Network Interface
31. Which of the following is a capability / term that only applies to a Layer 7 firewall?
- a. Packet filtering
 - b. Stateful inspection
 - c. Application gateway
 - d. Proxy-based firewall
32. Which of the following options for ipconfig will reset the contents of the DNS client resolver cache?
- a. ipconfig /cleardns
 - b. ipconfig /purgedns
 - c. ipconfig /flushdns
 - d. ipconfig /registerdns
33. Which of the following options for ipconfig will send a DHCPRELEASE message to the DHCP server to release the current DHCP configuration and discard the IP address configuration?
- a. ipconfig /release
 - b. ipconfig /purgedhcp
 - c. ipconfig /renew
 - d. ipconfig /cleardhcp

34. As a network administrator, you are asked to troubleshoot an intermittent connectivity issue. Which of the following commands will continue sending echo request messages to 172.16.1.254 until interrupted?
- ping 172.16.1.254
 - 172.16.1.254 ping
 - ping 172.16.1.254 /n -1
 - ping 172.16.1.254 /t
35. Which of the following issues can occur if there is a lack of documentation and all devices within a network have static IP address assignments?
- Duplicate IP addresses
 - Incorrect subnet mask
 - Incorrect default gateway
 - All of these are correct
36. As a network administrator, you are troubleshooting an issue in which you cannot connect to a router via telnet. The router is 10 hops away from the administrative client. Which of the following utilities can be used to identify where the connection is being dropped?
- Ping
 - ARP
 - Tracert
 - Ipconfig
37. A client machine is connected to switch port 23. The switch has learned the MAC address of the client NIC. If the user disconnects the NIC from the switch port, will the switch retain the MAC address?
- The MAC address will timeout in 300 seconds
 - The MAC address will be erased immediately
 - The MAC address will be retained in the MAC address table
 - None of these are correct
38. If a client has two default gateways, which would be taken for routing packets?
- The route with the highest metric value
 - The route with the lowest metric value
 - The first route assigned to a NIC with the highest IP address
 - The first route assigned to a NIC with the lowest IP address
39. Which of the following commands will show the DNS server IP address(es) on a Windows Server?
- ifconfig /all
 - ipconfig /all
 - ifconfig /showdns
 - ipconfig /showdns

40. What is the network IP range used for APIPA (Automatic Private IP Addressing)?
- 169.254.0.0/16
 - 192.168.1.1/24
 - 10.10.10.0/16
 - 127.0.0.1/16
41. Which of the following is also considered a Physical Address?
- MAC address (48-bit)
 - IP address (IPv4 / 32-bit and IPv6 / 128-bit)
 - Loopback
 - VLAN
42. As a network administrator, you are troubleshooting a client connectivity issue. You run ipconfig on the client and see that the IP address is 169.254.0.73. Which of the following is the most likely culprit?
- DNS is not working properly
 - VPN is configured
 - DHCP is unavailable
 - LBFO teaming is enabled
43. Which of the following is also considered a Logical Address?
- MAC address (48-bit)
 - IP address (IPv4 / 32-bit and IPv6 / 128-bit)
 - Loopback
 - VLAN
44. Which of the following steps / utilities would be most helpful for troubleshooting a DNS issue?
- Verify connectivity to the DNS server using ping
 - Verify name resolution using nslookup
 - Verify the DNS server settings using ipconfig
 - All of these are correct
45. Which of the following describes DHCP?
- DHCP is used to assign IP addresses to devices automatically
 - DHCP is necessary for accessing the gateway
 - DHCP is used to resolve hostnames to IP addresses
 - DHCP is necessary for wireless connectivity
46. Which of the following ranges is valid for IPv4 addresses?
- 10.0.0.0 - 10.255.255.255
 - 172.16.0.0 - 172.31.255.255
 - 192.168.0.0 - 192.168.255.255
 - All of these are correct

47. Which of the following OSI layers does ARP belong to?
- a. Layer 1 (Physical)
 - b. Layer 2 (Data Link)
 - c. Layer 3 (Network)
 - d. Layer 4 (Transport)
48. Which of the following commands will show a list of IP addresses and MAC addresses on a Windows client / server?
- a. ipconfig
 - b. show tech-support
 - c. arp -a
 - d. ifconfig
49. As a network administrator, you are troubleshooting within an environment that uses DHCP. The issue is that some clients are receiving IP addresses on the 192.168.1.0/24 network, while some are receiving IP addresses on the 10.10.10.0/24 network. Which of the following is the most likely culprit?
- a. Incorrect default gateway
 - b. DNS is misconfigured
 - c. Rogue DHCP server
 - d. Subnet mask is misconfigured
50. As a network administrator, you are troubleshooting an issue in which all clients lost internet connectivity after adding a new client to the network. The IP address assigned to the new client is 192.168.1.1/24. The clients regain internet connectivity if the new client is shut down, what is the most likely culprit?
- a. DNS is misconfigured
 - b. DHCP is misconfigured
 - c. IP address is not compatible with the provided subnet mask
 - d. IP address conflict with the default gateway